Project 2: Data Analysis with Pandas

Code:

Internship > Second Code > main.py > ...

```python
# --------------------------------------------
# Pandas Data Analysis Project
# Objective:
# 1. Load CSV dataset
# 2. Inspect data
# 3. Clean missing values
# 4. Filter records
# 5. Group and Aggregate data
# 6. Display simple insights
# --------------------------------------------

import pandas as pd
from pathlib import Path

# --------------------------------------------
# Load Dataset
# --------------------------------------------
csv_path = Path(__file__).with_name("students.csv")

try:
    df = pd.read_csv(csv_path)
except FileNotFoundError:
    print(f"Error: '{csv_path.name}' not found. Creating an empty DataFrame to proceed.")
    # Create an empty DataFrame with expected columns to prevent further errors.
    df = pd.DataFrame(columns=["ID", "Name", "Marks", "Age", "Department"])


print("========== Original Dataset ==========")
print(df)

# --------------------------------------------
# Inspect Dataset
# --------------------------------------------
print("\nDataset Information")
print(df.info())

print("\nMissing Values")
print(df.isnull().sum())

print("\nStatistical Summary")
print(df.describe())

# --------------------------------------------
# Clean Missing Data
# --------------------------------------------
for column in ["Marks", "Age"]:
    if column in df.columns:
        df[column] = pd.to_numeric(df[column], errors="coerce")
        if df[column].notna().any():
            df[column] = df[column].fillna(df[column].mean())

print("\n========== Cleaned Dataset ==========")
print(df)

```

```python

# --------------------------------------------
# Filtering
# --------------------------------------------
print("\nStudents Scoring Above 90")

top_students = df[df["Marks"] > 90]

print(top_students)

# --------------------------------------------
# Grouping and Aggregation
# --------------------------------------------
print("\nAverage Marks Department Wise")

department_marks = df.groupby("Department")["Marks"].mean()

print(department_marks)

print("\nMaximum Marks Department Wise")

print(df.groupby("Department")["Marks"].max())

print("\nNumber of Students in Each Department")

print(df.groupby("Department")["ID"].count())

# --------------------------------------------
# Highest Scorer
# --------------------------------------------
if not df.empty and "Marks" in df.columns and not df["Marks"].empty:
    # This block will only execute if there are students and marks data
    highest = df.loc[df["Marks"].idxmax()]
    print("\nHighest Scoring Student")
    print(highest)
else:
    # Handle the case where there are no students or no marks data to find a highest scorer
    print("\nNo students or marks data available to determine the highest scorer.")

print("\nProgram Completed Successfully!")
```

Internship > Second Code > students.csv > data

```
ID,Name,Department,Marks,Age
101,Jatin,AIML,95,19
102,Rahul,CSE,88,20
103,Aman,AIML,,19
104,Riya,CSE,91,20
105,Priya,IT,85,
106,Karan,AIML,78,21
107,Sneha,CSE,90,20
108,Arjun,IT,82,19
```